Module 01 – Project

Streamlit Tutorial

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Objectives

Basic Streamlit

- Introduction
- Setup & Installation
- Text Elements
- Media Elements
- Input Widgets
- File
- Form
- Session

Application

- Word Correction
- Object Detection
- Chatbot

Outline

SECTION 1
Basic Streamlit

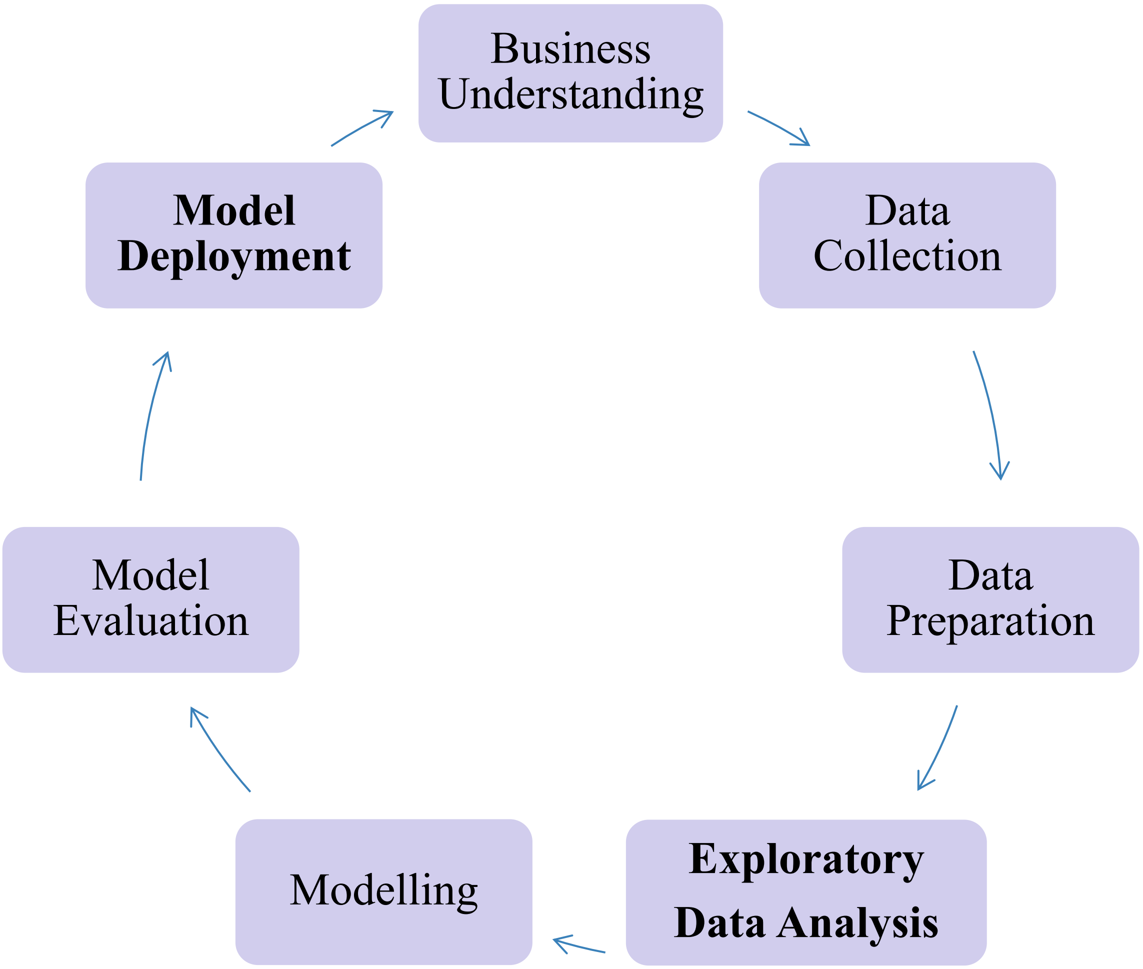
SECTION 2
Word Correction

SECTION 3
Object Detection

SECTION 4
Chatbot

Basic Streamlit

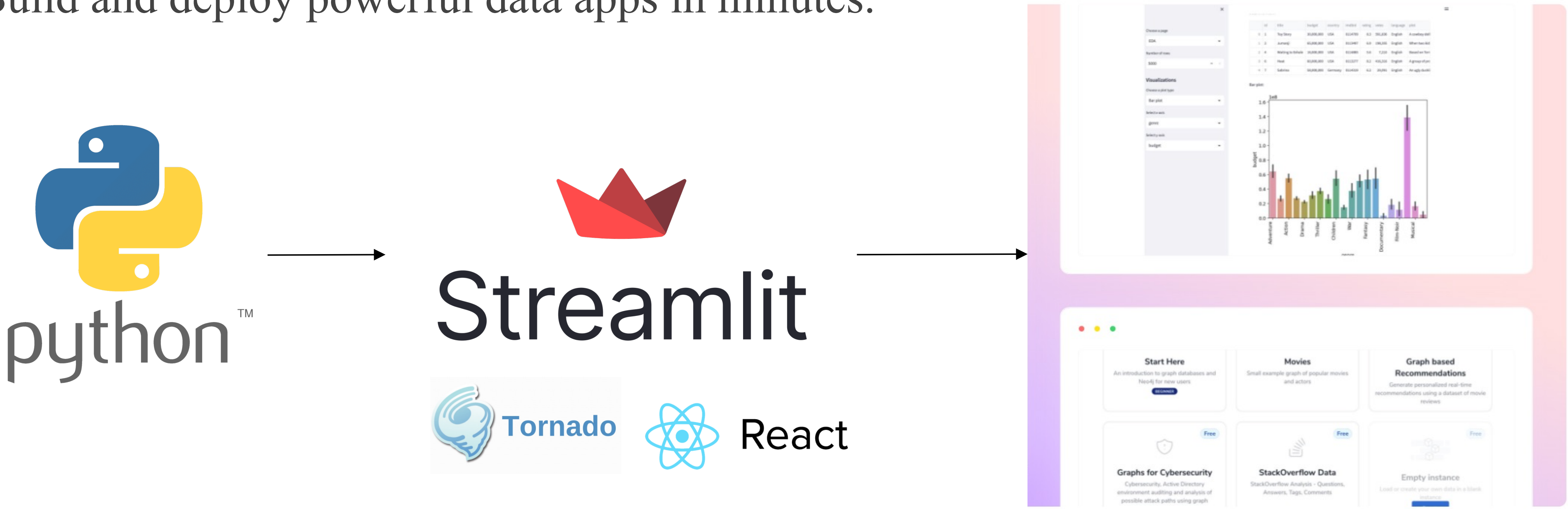
Why is Streamlit?



Basic Streamlit

What is Streamlit?

- Streamlit is an open-source Python framework for data scientists and AI/ML engineers to deliver dynamic data apps with only a few lines of code.
- Build and deploy powerful data apps in minutes.



Basic Streamlit

Setup & Installation

- Setup Virtual Environment

```
!conda create --name streamlit_env
!conda activate streamlit_env
```

- Installation

```
!pip install streamlit
!streamlit hello
!streamlit -version
!streamlit cache clear
>>> import streamlit as st
!streamlit run main.py
```

Basic Streamlit

Text Elements

- Display text

```python
import streamlit as st

st.title("MY PROJECT")
st.header("This is a header")
st.subheader("This is a subheader")
st.text("I love AI VIET NAM")
```

MY PROJECT

This is a header

This is a subheader

This is a caption

```
I love AI VIET NAM
```

```
Built with Strealit
```

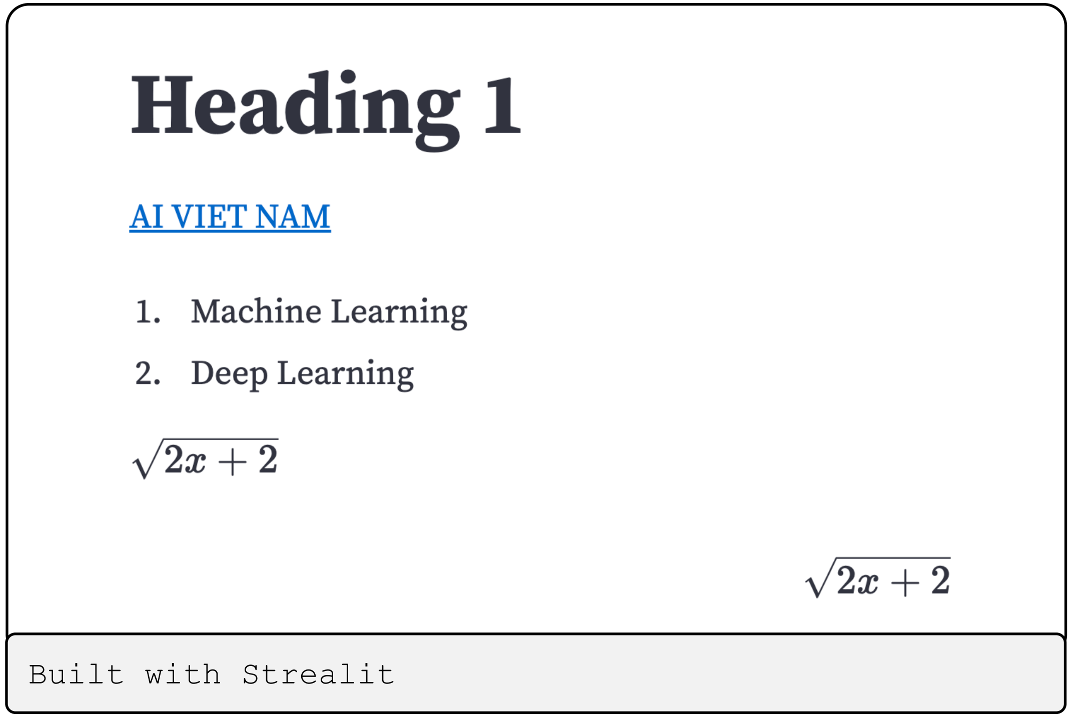
Basic Streamlit

Text Elements

- Display string formatted as Markdown

```python
import streamlit as st

st.markdown("# Heading 1")
st.markdown("[AI VIET 
NAM](https://aivietnam.edu.vn/)")
st.markdown("""
        1. Machine Learning
        2. Deep Learning
        """)
st.markdown("$\sqrt{2x+2}$")
st.latex("\sqrt{2x+2}")
```



AI VIET NAM
@aivietnam.edu.vn

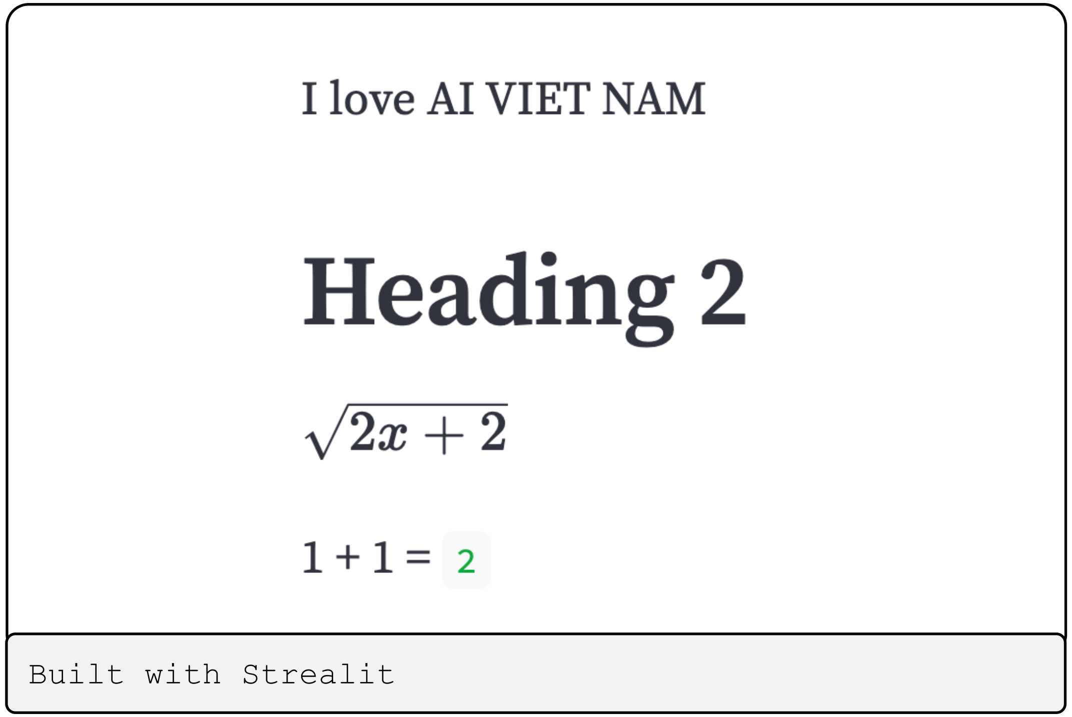
Basic Streamlit

Text Elements

- Write arguments to the app.

```
import streamlit as st

st.write('I love AI VIET NAM')
st.write('## Heading 2')
st.write('$ \sqrt{2x+2} $')
st.write('1 + 1 = ', 2)
```



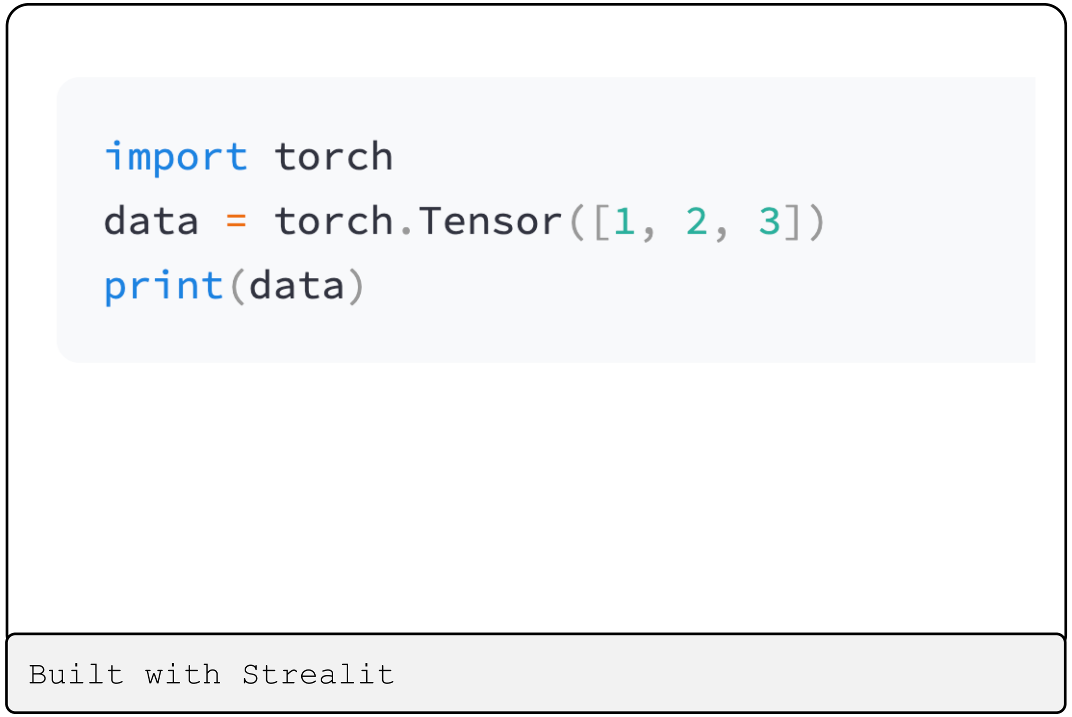
Basic Streamlit

Text Elements

- Code format.

```python
import streamlit as st

st.code("""
    import torch
    data = torch.Tensor([1, 2, 3])
    print(data)
""", language="python")
```



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Basic Streamlit

Text Elements

- Code format.

```python
import streamlit as st

def get_user_name():
    return 'Thai'
with st.echo():
    st.write('This code will be
    printed’)
    def get_email():
        return 'thai@gmail.com’
    user_name = get_user_name()
    email = get_email()
    st.write(user_name, email)
```

Basic Streamlit

Media Elements

- Logo

```python
import streamlit as st

st.logo('./logo.png')
```

Basic Streamlit

Media Elements

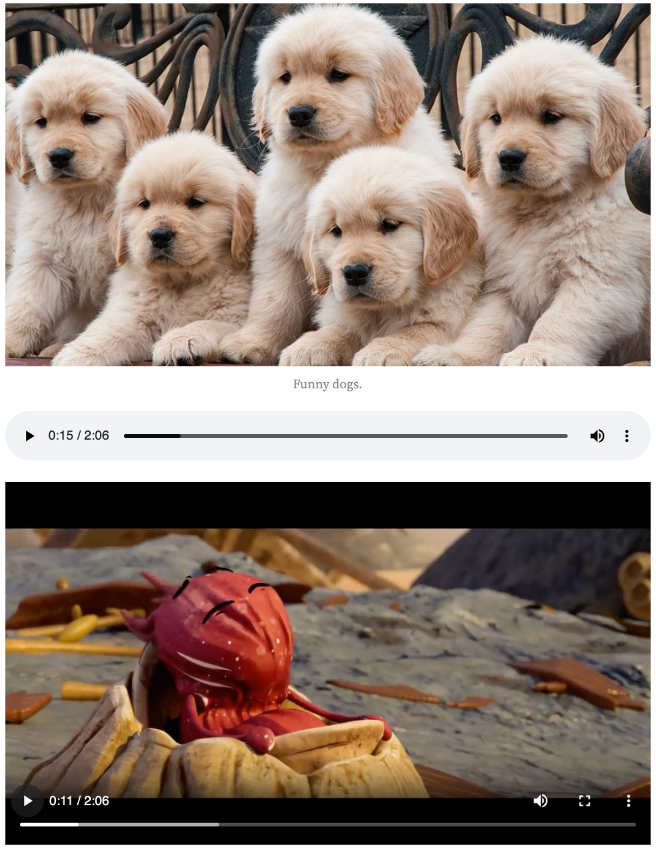
- Image – Audio - Video

```python
import streamlit as st

st.image(
    './dogs.jpeg',
    caption='Funny dogs.'
)

st.audio('./audio.mp4')

st.video('./video.mp4')
```



Built with Strealit

Basic Streamlit

Input Widgets

- Selection elements: Checkbox – Radio

```python
import streamlit as st

def get_name():
    st.write("Thai")
agree = st.checkbox("I  agree",
    on_change=get_name )
if agree:
    st.write("Great!")

st.radio(
    "Your favorite color:",
    ['Yellow', 'Bleu'],
    captions = ['Vàng', 'Xanh'])
```

Basic Streamlit

Input Widgets

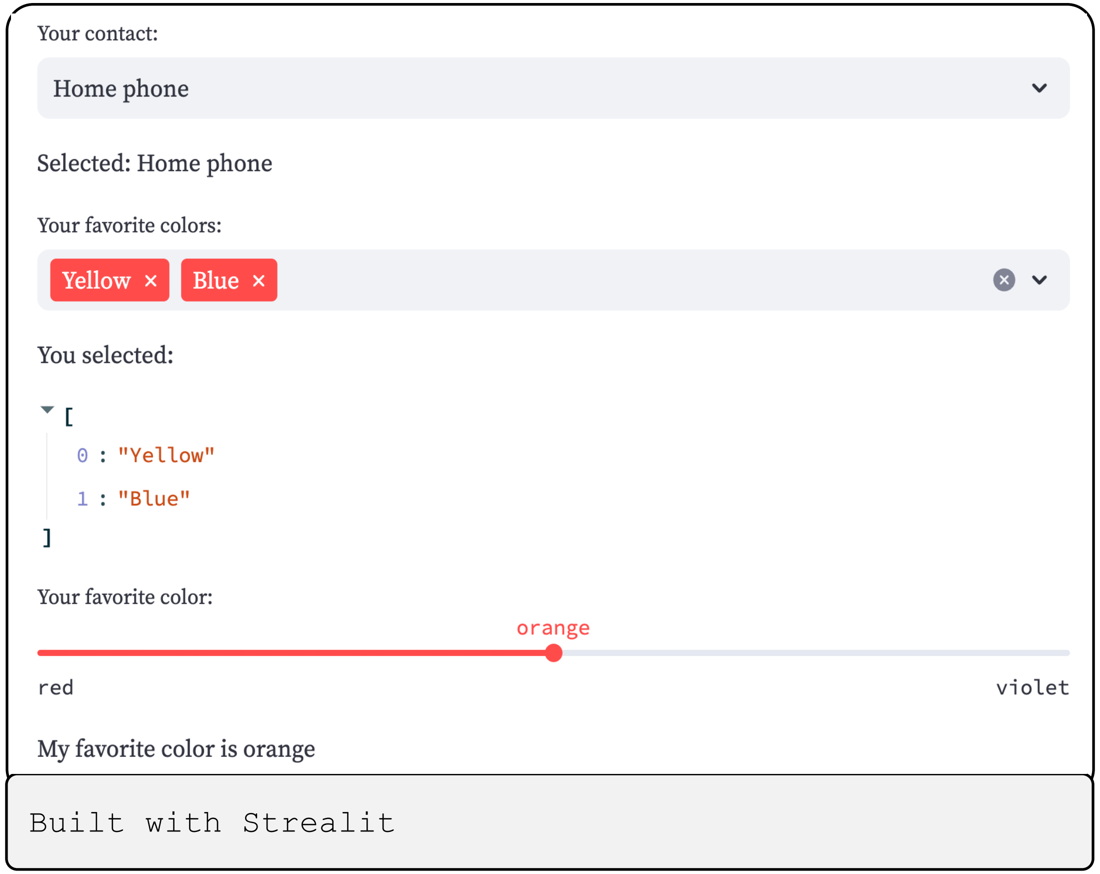
- Selection elements: Selectbox – Multiselect – Select slider

```python
import streamlit as st

option = st.selectbox(
"Your contact:",
("Email", "Home phone", "Mobile phone"))
st.write("Selected:", option)

options = st.multiselect(
"Your favorite colors:",
["Green", "Yellow", "Red", "Blue"],
["Yellow", "Red"])
st.write("You selected:", options)

color = st.select_slider(
"Your favorite color:",
options=["red", "orange", "violet"])
st.write("My favorite color is", color)
```



Basic Streamlit

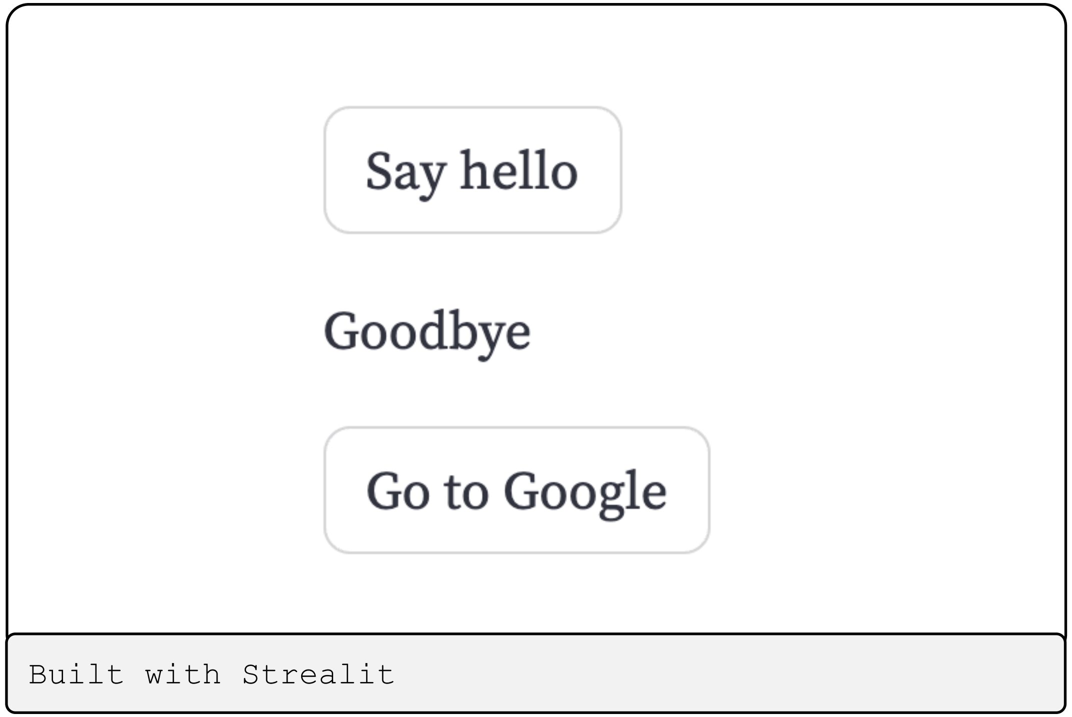
Input Widgets

- Button elements

```python
import streamlit as st

if st.button("Say hello"):
    st.write("Hello")
else:
    st.write("Goodbye")

st.link_button(
    "Go to Google",
    "https://www.google.com.vn/"
)
```



Basic Streamlit

Text Input Elements

- Text input

```python
import streamlit as st

title = st.text_input(
    "Movie title:", "Life of Brian"
)
st.write("The current movie title is", title)
```

Movie title:

Mai

The current movie title is Mai

Built with Strealit

Basic Streamlit

Text Input Elements

- Chat Elements

```python
import streamlit as st

messages = st.container(height=200)
if prompt := st.chat_input
    ("Say something"):
messages.chat_message("user")
    .write(prompt)
messages.chat_message("assistant")
    .write(f"Echo: {prompt}")
```

Basic Streamlit

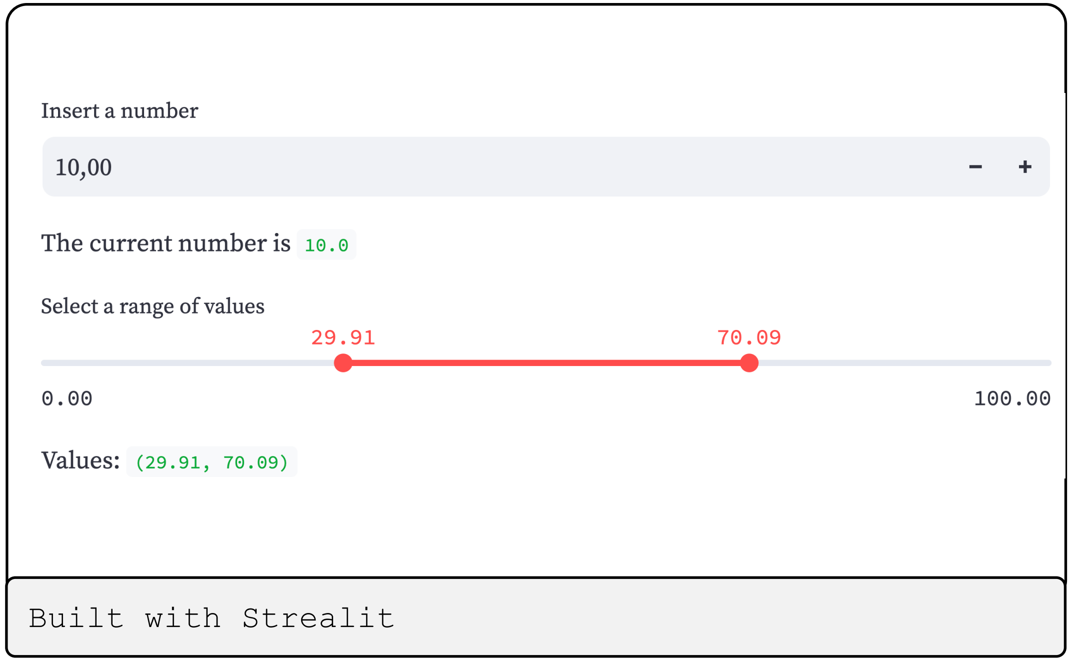
Text Input Elements

➢ Numeric input

```python
import streamlit as st

number = st.number_input("Insert a 
number")
st.write("The current number is ", 
number)

values = st.slider(
"Select a range of values", 
0.0, 100.0, (25.0, 75.0))
st.write("Values:", values)
```



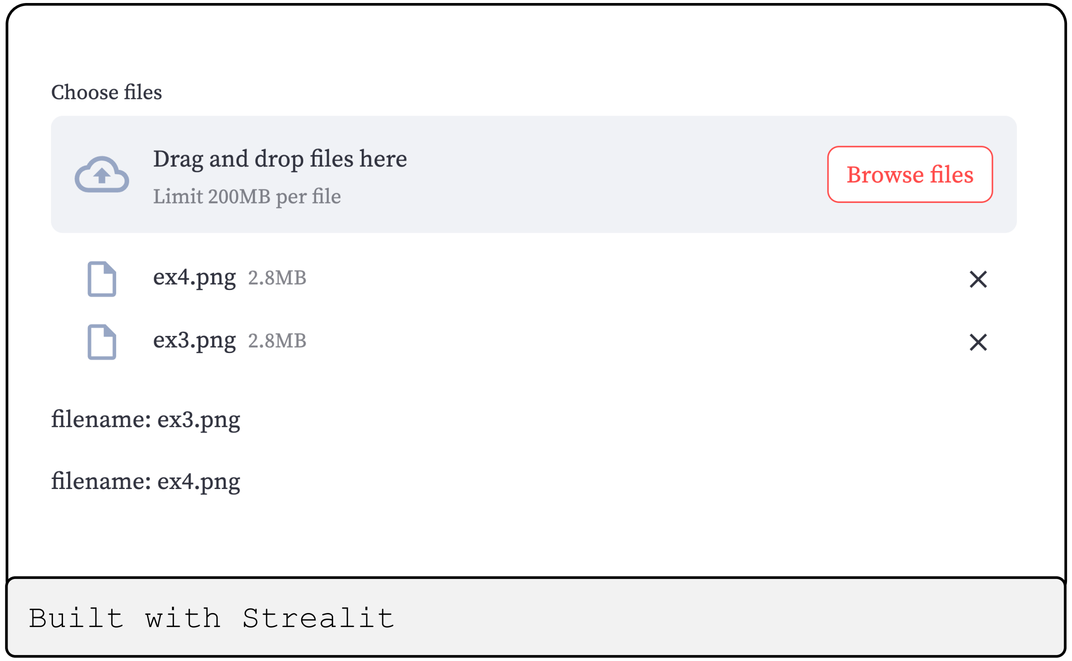
Basic Streamlit

File Uploader Widget

- Upload multiple files

```python
import streamlit as st

uploaded_files = st.file_uploader(
"Choose files",
accept_multiple_files=True)
for uploaded_file in uploaded_files:
    bytes_data = uploaded_file.read()
    st.write("filename:",
    uploaded_file.name)
```



Basic Streamlit

Form

- Create a form with a “Submit” button

```
import streamlit as st

with st.form("my_form"):
  col1, col2 = st.columns(2)
  f_name = col1.text_input('First Name’)
  l_name = col2.text_input('Last Name’)
  submitted = st.form_submit_button("Submit")
  if submitted:
     st.write("First Name: ", f_name,
     " - Last Name:", l_name)
```

Outline

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Basic Streamlit

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Word Correction

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Word Correction

Description

- Correct the spelling based on levenshtein distance.

levenshtein_distance(hel, hello) = 2

Word: hel

Build vocab

Vocab
1. hello
2. hi
3. love

Compute distances →

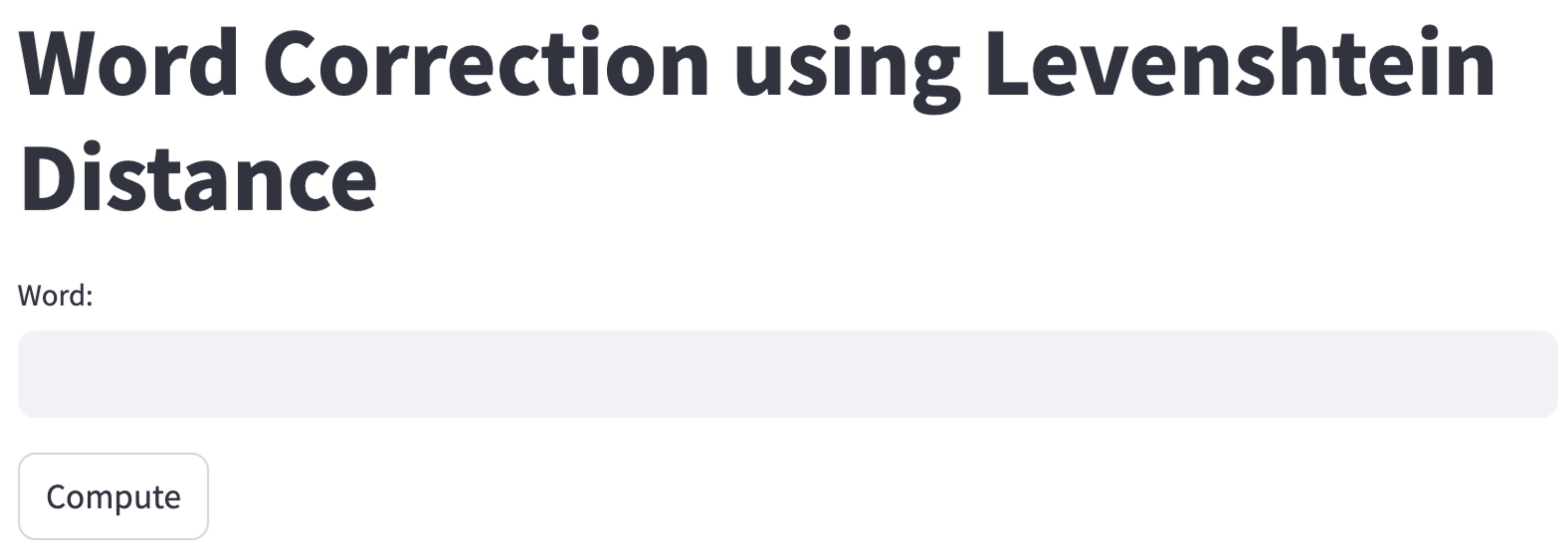
Distance
2
2
3

Correct: hello

Word Correction

Solution

- Using streamlit.



Word Correction

Solution

- Load the dictionary.

```
def load_vocab(file_path):
    with open(file_path, 'r') as f:
    lines = f.readlines()
    words = sorted(set([line.strip().lower() for line in lines]))
    return words

vocabs = load_vocab(file_path='./vocab.txt')
```

Word Correction

Solution

- Using streamlit.

```
def levenshtein_distance(word1, word2):
    return # distance

leven_distances = dict()
for vocab in vocabs:
    leven_distances[vocab] = levenshtein_distance(word, vocab)
# sorted by distance
sorted_distences = dict(sorted(leven_distances.items(), key=lambda item: item[1]))
correct_word = list(sorted_distences.keys())[0]
st.write('correct word: ', correct_word)
```

Word Correction

Result

Word:

bok

Compute

Correct word: book

Vocabulary:

```
[
  0 : "apple"
  1 : "book"
  2 : "dog"
  3 : "hello"
  4 : "never"
  5 : "please"
  6 : "random"
  7 : "sleep"
  8 : "start"
  9 : "understand"
]
```

Distances:

```
{
  "book" : 1
  "dog" : 2
  "apple" : 5
  "hello" : 5
  "never" : 5
  "random" : 5
  "sleep" : 5
  "start" : 5
  "please" : 6
  "understand" : 10
}
```

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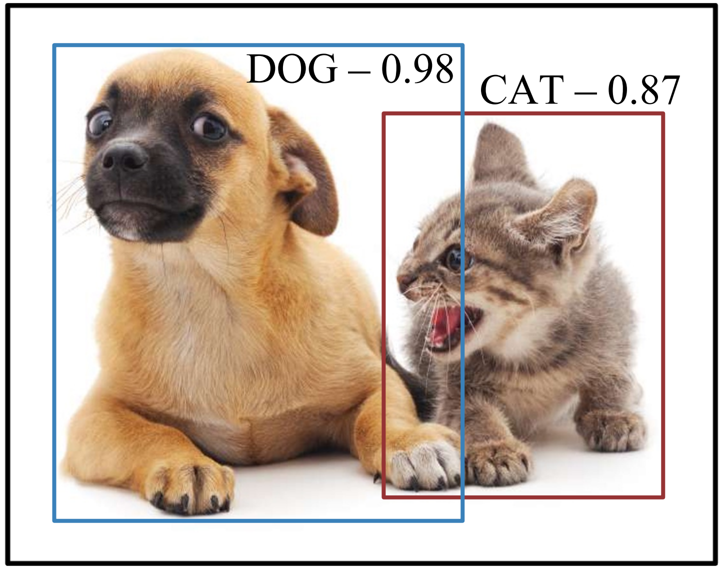
Object Detection

Description

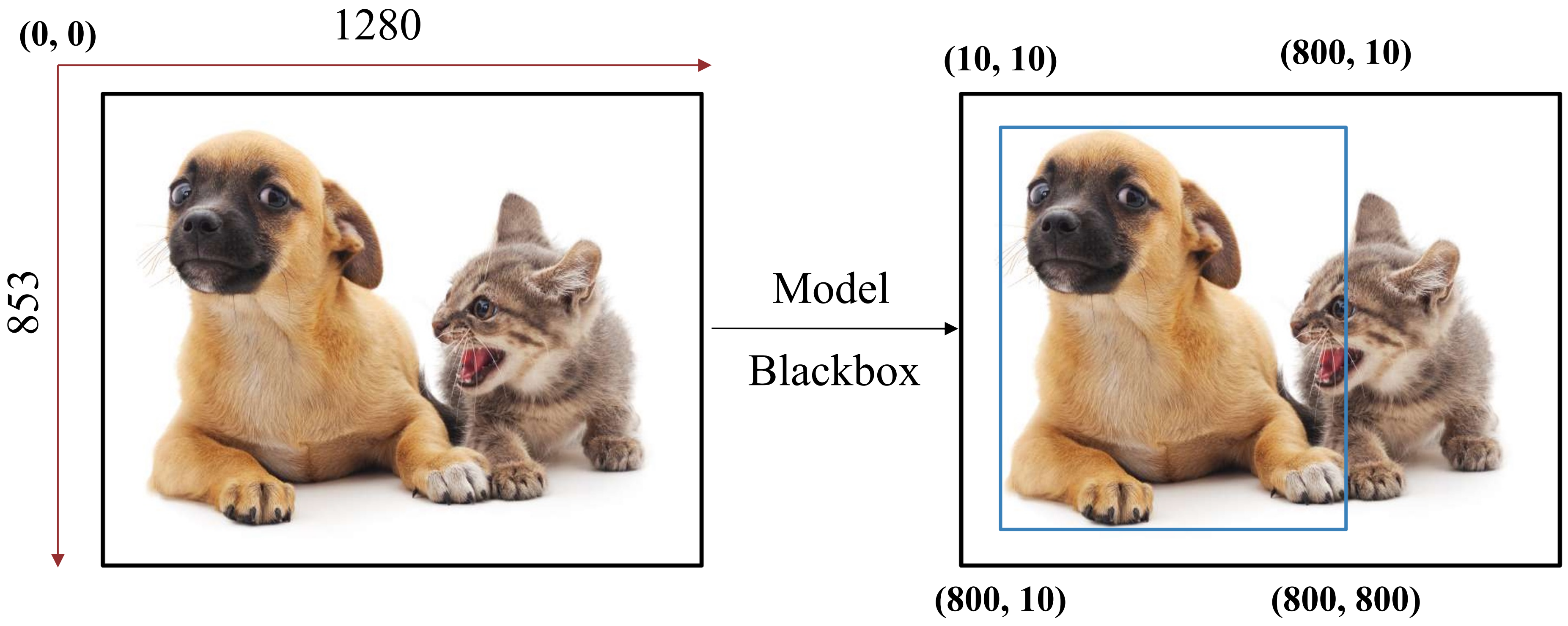
- Assign labels, bounding boxes to objects in the image



Object Detection

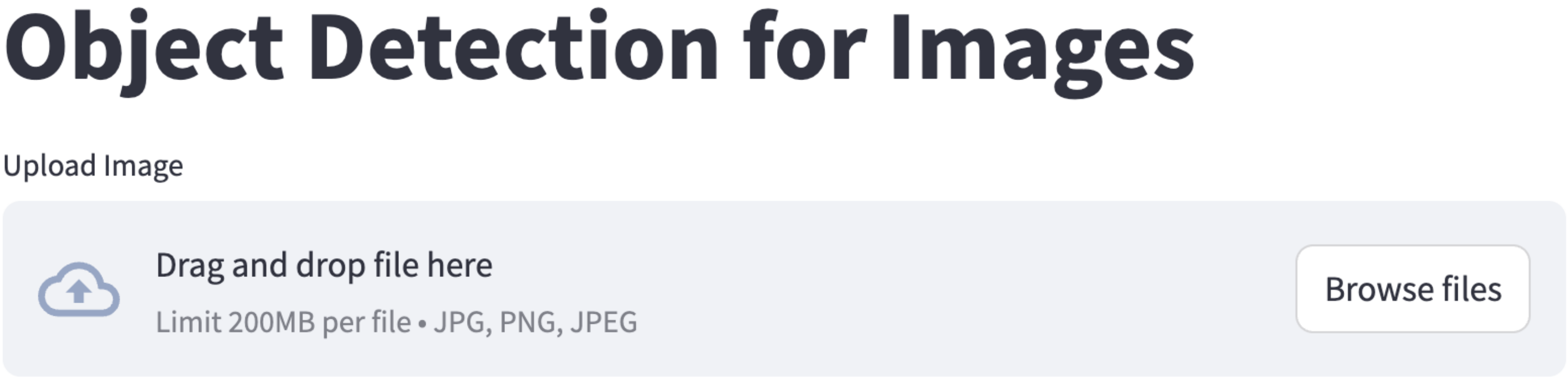
Description

- Assign labels, bounding boxes to objects in the image



Object Detection

UI




```python
st.title('Object Detection for Images')
file = st.file_uploader('Upload Image', type = ['jpg','png','jpeg'])
if file is not None:
    st.image(file, caption = "Uploaded Image")
```

Object Detection

Detection using opencv

```
def process_image(image):
    blob = cv2.dnn.blobFromImage(
        cv2.resize(image, (300, 300)), 0.007843, (300, 300), 127.5
    )
    net = cv2.dnn.readNetFromCaffe(PROTOTXT, MODEL)
    net.setInput(blob)
    detections = net.forward()
    return detections
```

Object Detection

Filter & Display

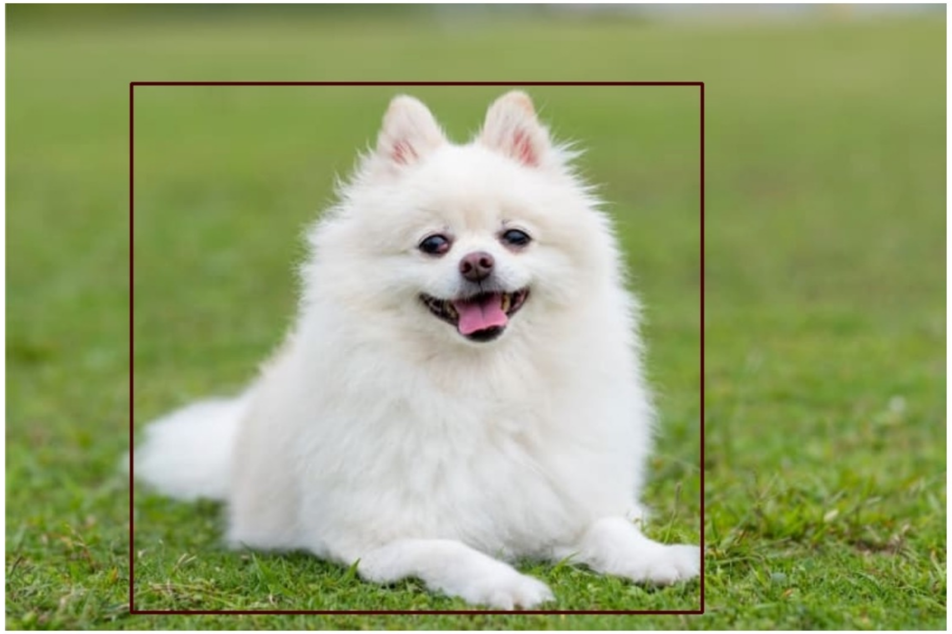
```
def annotate_image(
        image, detections, confidence_threshold=0.5
    ):
    # loop over the detections
    (h, w) = image.shape[:2]
    for i in np.arange(0, detections.shape[2]):
        confidence = detections[0, 0, i, 2]

        if confidence > confidence_threshold:
        # extract the index of the class label from the `detections`,
        # then compute the (x, y)-coordinates of the bounding box for
        # the object
        idx = int(detections[0, 0, i, 1])
        box = detections[0, 0, i, 3:7] * np.array([w, h, w, h])
        (startX, startY, endX, endY) = box.astype("int")
        cv2.rectangle(image, (startX, startY), (endX, endY), 70, 2)
    return image
```

Object Detection

Result

Uploaded Image



Processed Image

Outline

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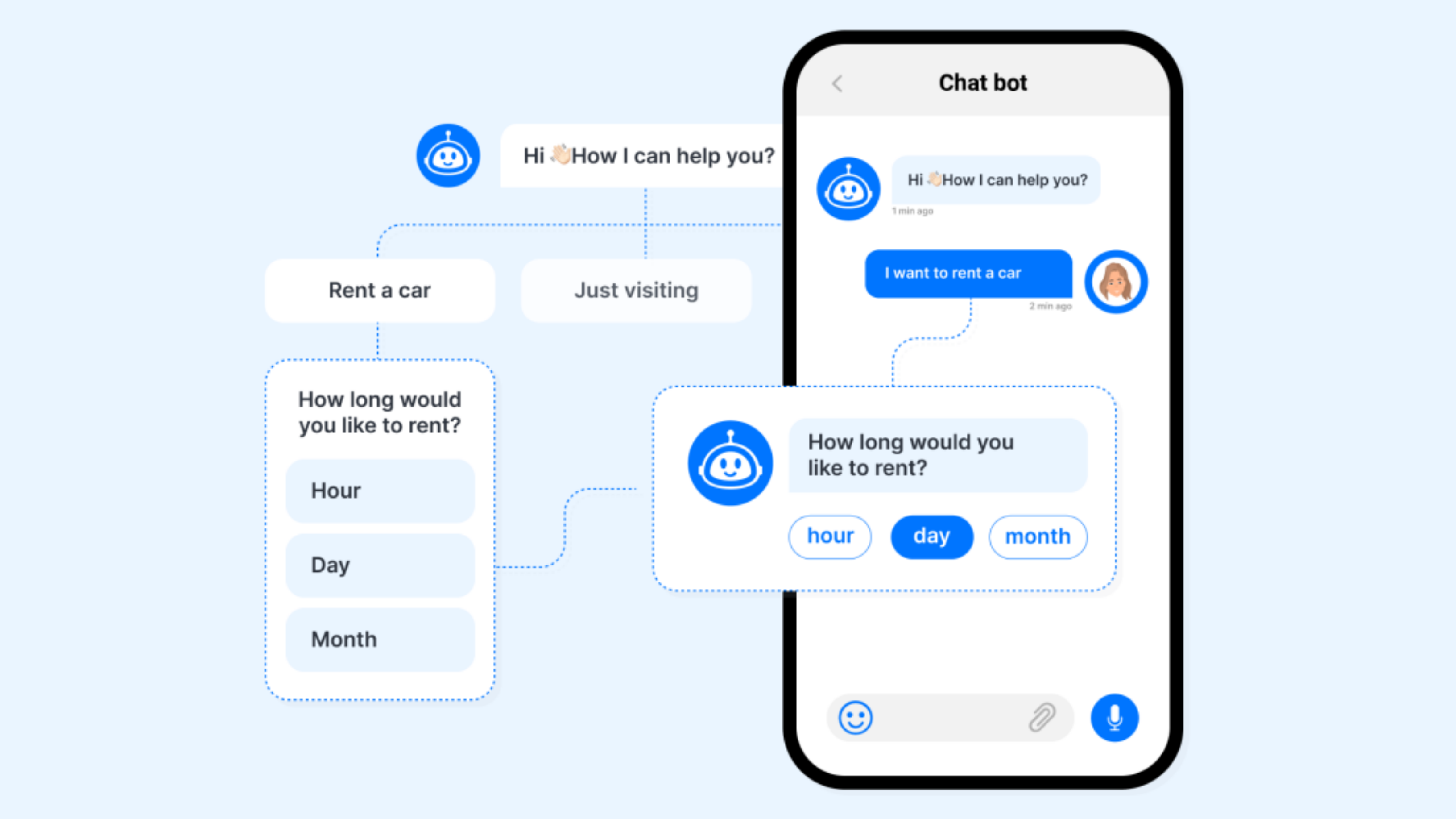
Chatbot

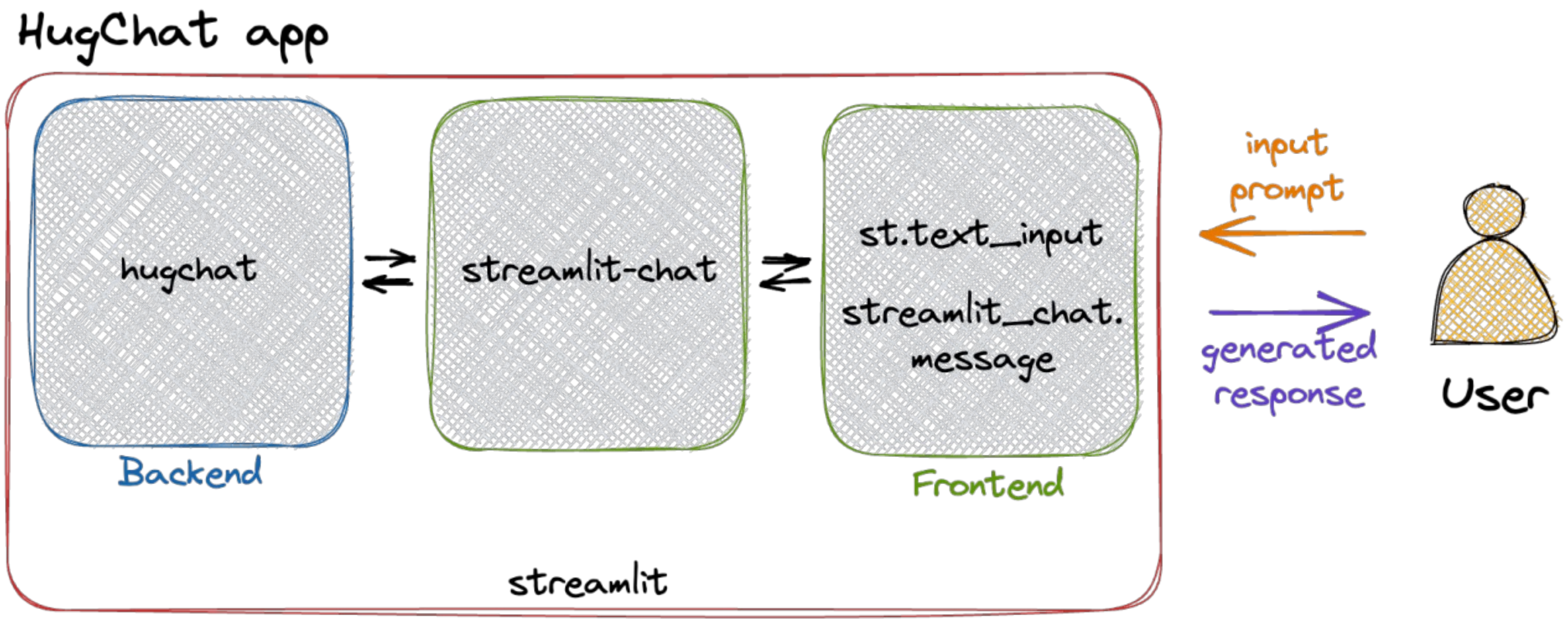
Description



Chatbot

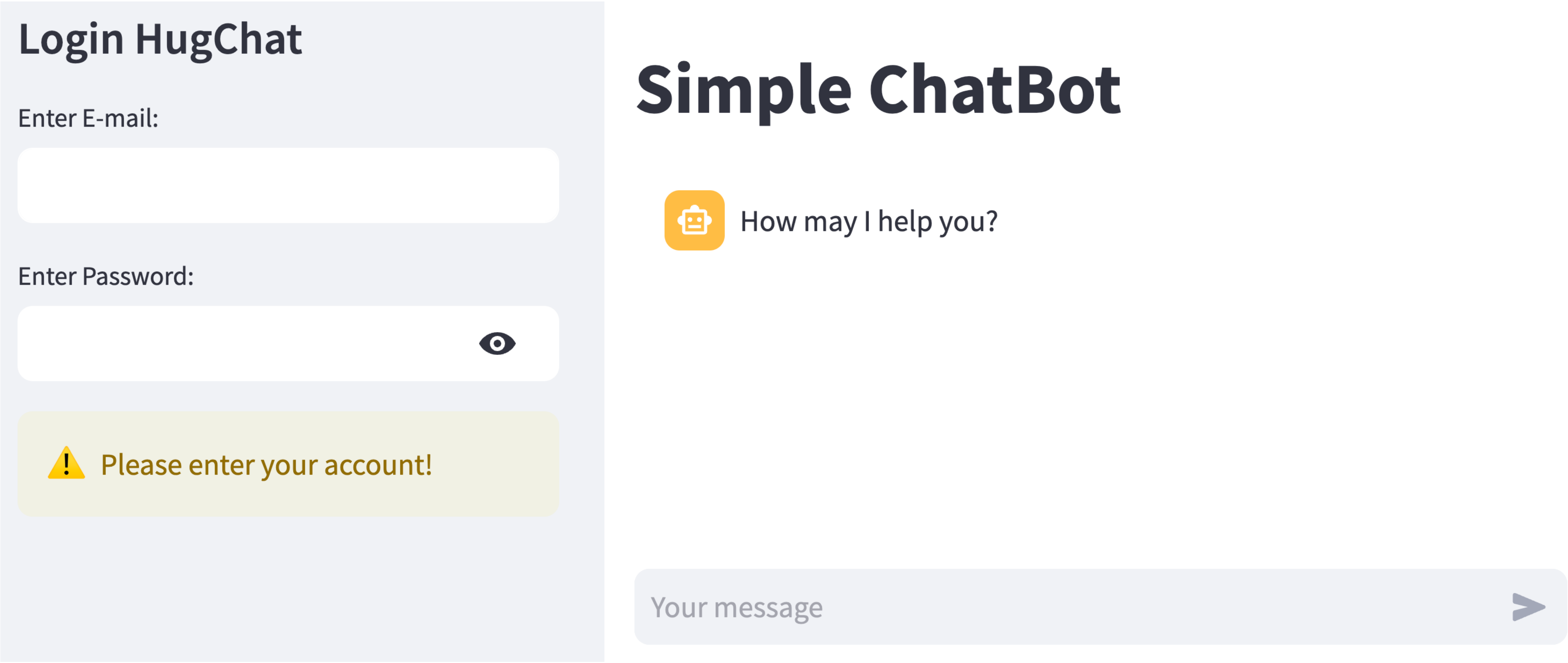
HugChat

- Hugging Chat Python API (Use Hugging Face Account).



Chatbot

UI


Login HugChat
Enter E-mail:
Enter Password:
Please enter your account!
Simple ChatBot
How may I help you?
Your message

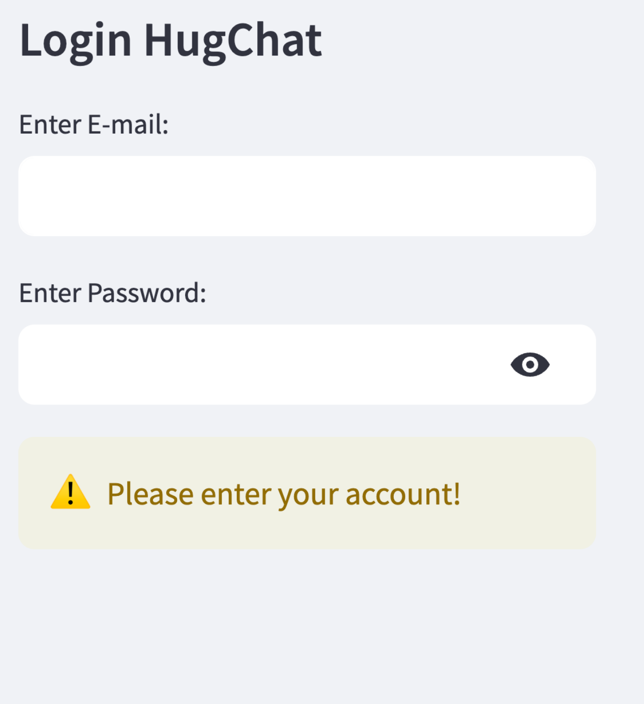
Chatbot

UI

- Hugging Face Account

```python
with st.sidebar:
    st.title('Login HugChat')
    hf_email = st.text_input('Enter E-mail:')
    hf_pass = st.text_input('Enter Password:', type='password')

    if not (hf_email and hf_pass):
        st.warning('Please enter your account!', icon='⚠️')
    else:
        st.success('Proceed to entering your prompt message!',
        icon='👉')
```



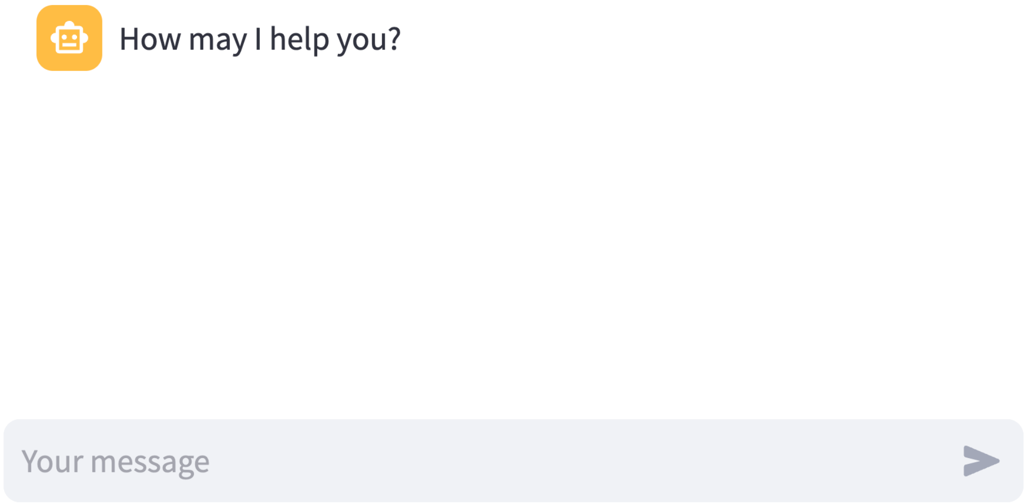
Chatbot

UI

- Display chat messages

```
# Store LLM generated responses
if "messages" not in st.session_state.keys():
    st.session_state.messages = [{"role": "assistant", "content": "How may I help you?"}]

# Display chat messages
for message in st.session_state.messages:
    with st.chat_message(message["role"]):
        st.write(message["content"])
```



Chatbot

UI

➤ Get response from hugchat

```
# Function for generating LLM response
def generate_response(prompt_input, email, passwd):
    # Hugging Face Login
    sign = Login(email, passwd)
    cookies = sign.login()
    # Create ChatBot
    chatbot = hugchat.ChatBot(cookies=cookies.get_dict())
    return chatbot.chat(prompt_input)
```

Chatbot

UI

- Generate a new response

```
# Generate a new response if last message is not from assistant
if st.session_state.messages[-1]["role"] != "assistant":
    with st.chat_message("assistant"):
        with st.spinner("Thinking..."):
            response = generate_response(prompt, hf_email, hf_pass)
            st.write(response)
    message = {"role": "assistant", "content": response}
    st.session_state.messages.append(message)
```

Chatbot

Result

Login HugChat

Enter E-mail:

Enter Password:

••••••••••••••••••••

👉 Proceed to entering your prompt message!

How may I help you?

Hi!

`Hi! How can I help you today?`

What is the capital of Vietnam

`The capital of Vietnam is Hanoi.`

Bạn có hiểu tiếng việt không?

`Vâng, tôi có thể hiểu một chút tiếng Việt. Tôi có thể giúp bạn bằng tiếng Việt nếu bạn muốn.`

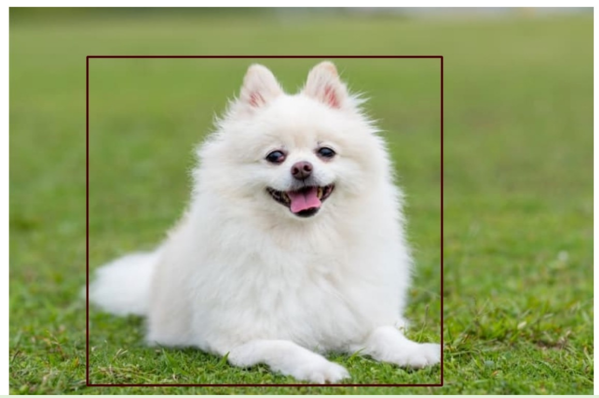
Your message

Summary

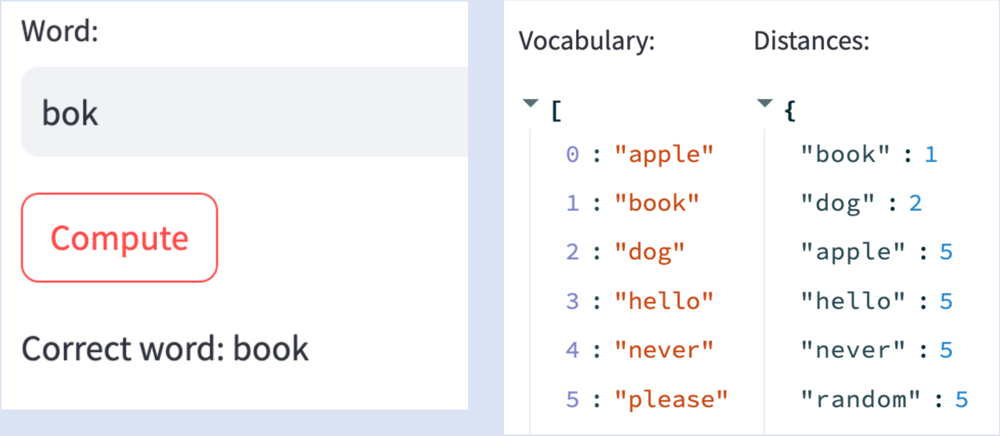
Basic Streamlit

```
!pip install streamlit
!streamlit hello
!streamlit -version
!streamlit cache clear
!streamlit run main.py
>>> import streamlit as st
>>> st.write(), st.text_input()
>>> st.file_uploader()
>>> st.sesstion_state()
```

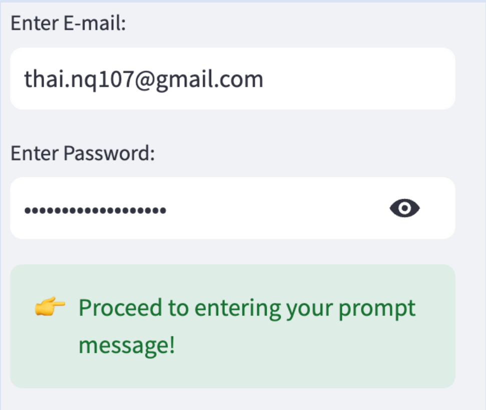
Object Detection

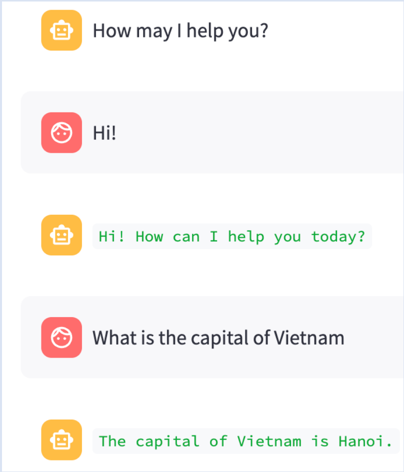


Word Correction



Chatbot





Thanks!

Any questions?